



Vidyavardhaka Sangha[®], Mysore
VIDYAVARDHAKA COLLEGE OF ENGINEERING

Autonomous Institute, Affiliated to Visvesvaraya Technological University, Belagavi
(Approved by AICTE, New Delhi & Government of Karnataka)

Accredited by NBA | NAAC with 'A' Grade
Department of Computer Science & Engineering

Phone: +91 821-4276230, Email: hodes@vvce.ac.in
Web: <http://www.vvce.ac.in>



@vvceofficial

USN 4VV22CS189

CONTINUOUS COMPREHENSIVE EVALUATION: TEST-1

COURSE NAME : Blockchain Technology		ACADEMIC YEAR : 2024-2025	
COURSE CODE : BCSBT603		DATE : 21/2/2025	
SEM / BRANCH : VI/CSE		MAX. MARKS : 15	
FACULTY : Anil Kumar B H, Lakshmi B S, Abdul Latif Saleem		DURATION : 1.0 Hrs.	
Q No.	Instructions to students: Answer one full question	Marks Allotted	BL CO/PO Mapping
MODULE 1			
1 a)	Explain the concepts of Distributed Systems and CAP Theorem in Blockchain.	8	L2 CO1/PO1
1 b)	Explain network view of Blockchain and structure of block with a neat diagram.	7	L2 CO1/PO1
OR			
2 a)	Explain the different features of Blockchain in detail.	8	L2 CO1/PO1
2 b)	Explain the requirements which must be met to provide the desired results in a consensus mechanism.	7	L2 CO1/PO1

COURSE OUTCOMES

CO1	Understand the blockchain terminologies with its applications.
CO2	Examine and apply the decentralization, asymmetric cryptographic primitives, and Bitcoin concepts
CO3	Analyze the principles of Ethereum and its transactions in blockchain
CO4	Design and demonstrate a blockchain using Ethereum Tool and make effective presentation in a team



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CONTINUOUS COMPREHENSIVE EVALUATION: TEST-2

COURSE NAME : Blockchain Technology	ACADEMIC YEAR : 2024-2025
COURSE CODE : BCSBT603	DATE : 02/4/2025
SEM / BRANCH : VI/CSE	MAX. MARKS : 30
FACULTY : Anil Kumar B H, Lakshmi B S, Abdul Latif Saleem	DURATION : 1.0 Hrs.

Q No.	Instructions to students: Answer full question	Marks Allotted	BL	CO/PO Mapping
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MODULE II

1 a)	Explain centralized and decentralized system with a neat diagram.	4	L2	CO1/PO1
1 b)	Illustrate the various steps of AES algorithm along with a neat diagram.	5	L3	CO2/PO1
1 b)	Apply the concepts of full ecosystem decentralization in order to achieve complete decentralization.	6	L3	CO2/PO1

MODULE III

2 a)	Apply RSA public-key encryption algorithm, for plaintext 3 for a given prime number 7 and 11 to generate the public and private keys	5	L3	CO2/PO1
2 b)	Apply SHA-256 to create fixed length digest used in blockchain.	5	L3	CO2/PO1
2 C)	Consider doctor's digital prescription and analyze how RSA digital signature will ensure authenticity and security by justifying with a neat diagram.	5	L4	CO3/PO2

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CONTINUOUS COMPREHENSIVE EVALUATION: TEST-3

COURSE NAME : Blockchain Technology	ACADEMIC YEAR : 2024-2025
COURSE CODE : BCSBT603	DATE : 16/5/2025
SEM / BRANCH : VI/CSE	MAX. MARKS : 30
FACULTY : Anil Kumar B H, Lakshmi B S, Abdul Latif Saleem	DURATION : 1.0 Hrs.

Q No.	Instructions to students: Answer one full question	Marks Allotted	BL	CO/PO Mapping
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MODULE 4

1 a)	Define wallet and explain any one type of wallet.	3	L2	CO1/PO1
1 b)	Analyze how the given instruction is executed using P2PKH script execution. scriptPubkey :OP_DUP OP_HASH160<pubHash> OP_EQUALVERIFY OP_CHECKSIG ScriptSig : <sig> <pubkey>	6	L4	CO3/PO2
1 c)	Analyze word of law, world of cryptography and world of Accountancy are integrated in Ricardian contract of bowtie model with a neat diagram.	6	L4	CO3/PO2

MODULE 5

2 a)	Illustrate the working principle of Ethereum virtual machine (EVM) with a neat diagram.	8	L3	CO2/PO1
2 b)	Analyze how the relationship exists between the transaction, transaction trie and Block header with relationship diagram.	7	L4	CO3/PO2

[OR]

3 a)	Illustrate the relationship between the accounts trie , account tuple, world state trie and state root hash with a neat diagram.	8	L3	CO2/PO1
3 b)	Analyze the given Data and evaluate the total gas cost for a transaction in Ethereum blockchain. Data "SHA3 has gas cost of 30 and current gas price is 25 GWei".	7	L4	CO3/PO2

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